Simulation Analysis

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1 How to read this document

This document is divided into sections based on specific topics. In each section there is a set of summary main bullet points, a description of the analysis done in that section to support those bullet points, and the output from the analysis done in that section. The main bullet points cover a combination of answering the questions in the empirical paper and the overall questions about how each of the estimators perform under different conditions. The analysis descriptions need to be moved to be under the piece of analysis they describe, but for now I thought it was better to have it all up top.

2 Research Questions

3 TO DO EDIT THIS BASED On The Revised Research Questions Doc

- Are any of the estimators ever substantially biased? When?
- Are any of the estimators generally more precise than others? When?
- Which estimators produce the best estimates of the treatment effect standard error?
- Which estimators produce the best estimates of the treatment effect standard error?
- Are the standard errors targeting finite versus super population

Relevant section from the empirical paper:

We discuss this further in our simulations, but the consequence is that any of our standard error estimators can be viewed as conservative, if we view the clusters as a finite population, or as appropriate if we consider the clusters as coming from a larger superpopulation (Litschwartz et al., 2025).

- Do estimated standard errors vary across estimators and why?

Notes: Is this because true SEs vary across estimators? Is this estimation error? Are certain SE estimators more accurate than others?

Relevant section from the empirical paper: Choosing an estimand and excluding the fully interacted estimators results in estimated standards errors that are less variable across estimators. However, it is notable that differences are still often non-trivial. Even in Figure we see estimated standard errors are 20 percent different across estimators in 24 percent of the cases we examined.

- Are the fully-interacted models driving the instability of the estimated standard errors?

Relevant section from the empirical paper:

For example, the largest estimated standard error is 1.5x the smallest estimated standard error in only 10 percent of cases without the fully-interacted models, down from 44 percent when considering all estimators. The ratio never exceeds 1.8 once the fully-interacted models are excluded. This corroborates simulation findings that show the instability of estimated standard errors coming from fully-interacted estimators (Litschwartz et al., 202X). In general, we do not recommend using the fully-interacted estimators due to this short-coming.

- Are problems caused by low degrees of freedom due to estimate instability or bias?

Relevant section from the empirical paper: Notice that in Figure 2 it appears that choice of estimator matters more in low-degree of freedom studies – the ranges across estimators tend to be larger in low degree of freedom studies (blue dots) than high degree of freedom studies (red dots).

[ADD SOMETHING ABOUT WHAT WE LEARNED FROM THE SIMULATIONS – DO WE THINK THESE RANGES ARE MORE ABOUT BIAS OR ESTIMATION ERROR]

4 Data Generating Process

The simulations are a fully factored experiment with the following data generating parameters.

Table 1: Simulation Data Generating Parameters

Parameter	**Values**
Block-Level Fixed	
Number of Blocks	5
Average Treatment Effect	.2
Block-Level ICC	.2
Average Number of Observations Per Cluster	20
Number of Block Level Covariates	0
Block-Level Varying	
Block Type	Same, Messy, Reasonable
Average Number of Clusters per Block	5, 10, 20
Proportion of Clusters Treated	.5, .7
Cluster-Level Varying	
Correlation of Impact and Size of Cluster	0, .8
Cluster Level Treatment Effect Standard Deviation	0, .3
Cluster Level ICC	0, .2
How much does the size of clusters vary (coefficient of variation)?	0, .33, .6
Number of Cluster-Level Covariates	0, 1, 5 [R^2 Always .5]

Table 2: Block-Type Parameters

Parameter	Blocks All The Same	Blocks All Messy	Blocks All Reasonable
How much do number clusters per block vary (coefficient of variation)?	0 (none)	.85 (big)	.33 (average)
Block-Level Average Treatment Effect Standard Deviation	0	.6	.1
Correlation Between Treatment Effect and Cluster Size	0	.8	.2
Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters	0	.8	.1
Correlation of Proportion of Clusters Treated per Block and Impact	0	.8	0
Correlation Between Outcome Intercept and Cluster Size	0	.8	.1

5 Are any of the estimators ever substantially biased? When?

In 90% of the scenarios we tested for all of the estimators the magnitude of the bias was less than .01 (Figure 1), and the scenarios with bias require the confluence of multiple bias causing factors (Figure 2). For clarity we just present the results from the full factored simulation, but results from the low degrees of freedom results showed the same patterns. The estimators theory says are unbiased do indeed have nearly no bias (or have very low bias) across the full range of simulations. The person weighted fully interacted did estimator had bias as bad as -.033 in the worst 5% of scenarios tested, which from the regression tree requires a correlation between the treatment effect and cluster size and a small number of clusters per block (Figure 2). For the remaining estimators, except the multi-level fully interacted model, the bias was as bad as -.2, which was primarily driven by the combination of dissimilar blocks and unbalanced treatment assignment of the clusters. The fully interacted multi-level model performed fine with dissimilar blocks and unbalanced treatment assignment, and overall only bias as bad as .1. However, all the multi-level models had bias in the simulations where there was a correlation between the number of clusters and the treatment effect, and the cluster-level ICC was zero, and there was large within block variance in cluster size.

5.1 TO DO

- I know the tree titles are weird, I'm going to fix it
- In general I need to clean up the graphs

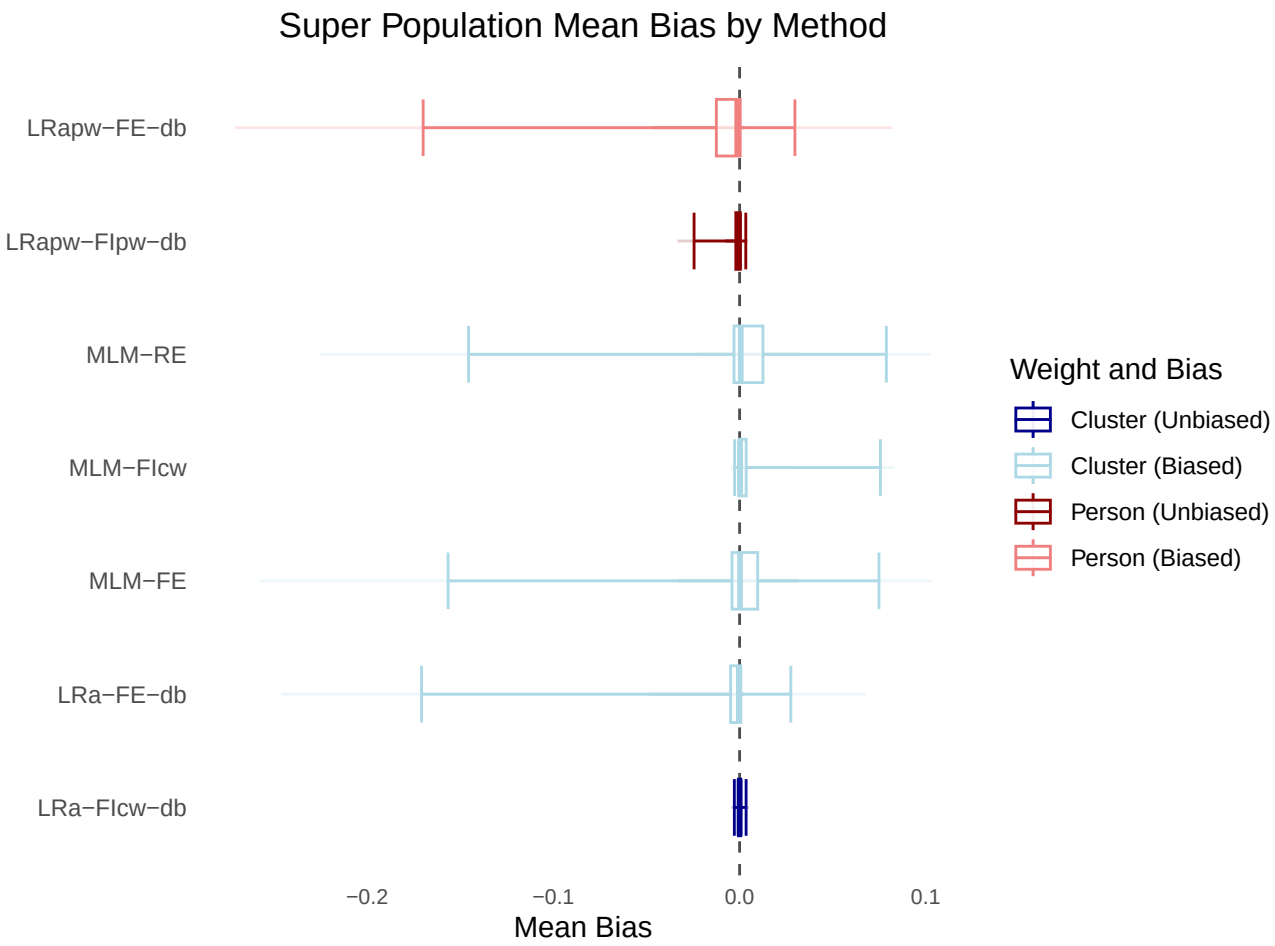
5.2 Main Bullet Points:

-
- All analysis done on the fullfactor simulation, because the results didn't change in the low degrees of freedom simulation

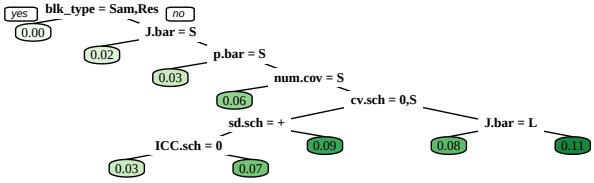
- Methods that are as bad as -.2 in the 5% of scenarios with the most bias: LRapw-FE-db, MLM-RE, MLM−FE, LRa−FE−db
- LRapw−FE−db: In the worst 5% of the sims bias as big as -.2. Fine as long as blocks aren't messy or aren't unbalanced or J.bar is S. If unbalanced but J.Bar is S and also cor_J_tau is not zero, that can be a problem.
- MLM−FE, mostly no bias when blocks aren't messy, except cor_J_tau != 0 & ICC.sch=0 & cv.sch!=0, when blocks are messy than bias is at J.bar M,L and p.bar is unbalanced
- LRa−FE−db: In the worst 5% of the sims bias as big as -.2. As long as the blocks are not messy it's fine. Bias needs combination J.Bar is Large and unbalance, otherwise it's fine.
- MLM−FIcw has bias as bad as .1 in the 5% of scenarios with most bias, bias is when cor_J_tau != 0 & ICC.sch =0 & cv.sch!=S
- The LRapw−FIpw−db . This is consistent with theory.
- LRa−FIcw−db: no bias in any case
- MLM-RE, mostly no bias when blocks aren't messy, except cor_J_tau != 0 & ICC.sch=0 & cv.sch!=0, when blocks are messy than bias is at J.bar M,L and p.bar is unbalanced
- For most of the estimators that are biased, the mean bias is mostly driven by large differences across blocks. With the largest bias in cases with large differences across blocks, 10 or 20 clusters, and unbalanced assignment to treatment (mean p =.7).

5.3 Description of Analysis

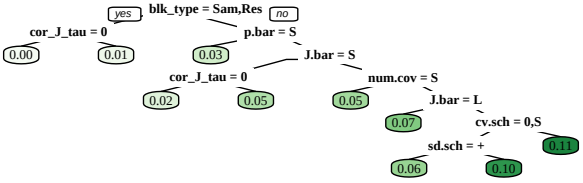
- The first graph is a boxplot of mean bias across the different scenarios by method. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.
- The second graph is a regression tree which predicts the magnitude of the mean bias using different data generating parameters and also flags for if the method is an MLM, is fully interacted, is cluster weighted, or is person weighted.
- The third graph is a the same regression tree above, but just for the LRapw−FIpw−db. The indicators for MLM, FI, cluster weighted, and person weighted are dropped for this regression.
- The first table contains the DGPs that produce that largest magnitude of mean bias for each method, conditional on the magnitude of the mean bias being at least .1 . Where again mean bias is the mean bias against the targeted estimator.


Super Population Mean Bias by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
−0.2
−0.1
0.0
0.1
Mean Bias
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

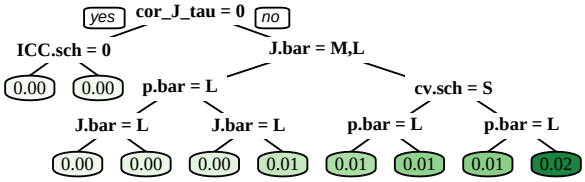
LRa−FE−db



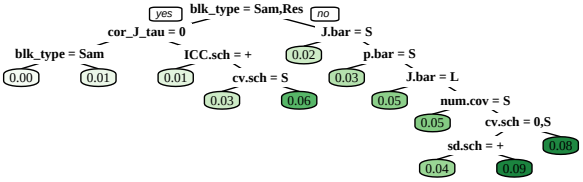
LRapw−FE−db



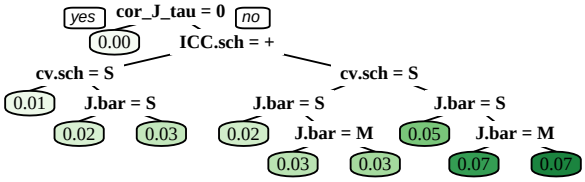
LRapw−FIpw−db



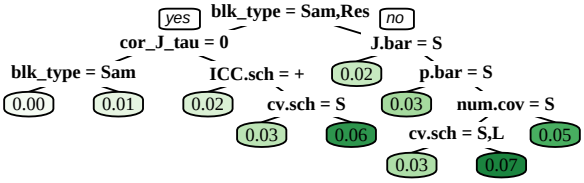
MLM−FE



MLM−FIcw

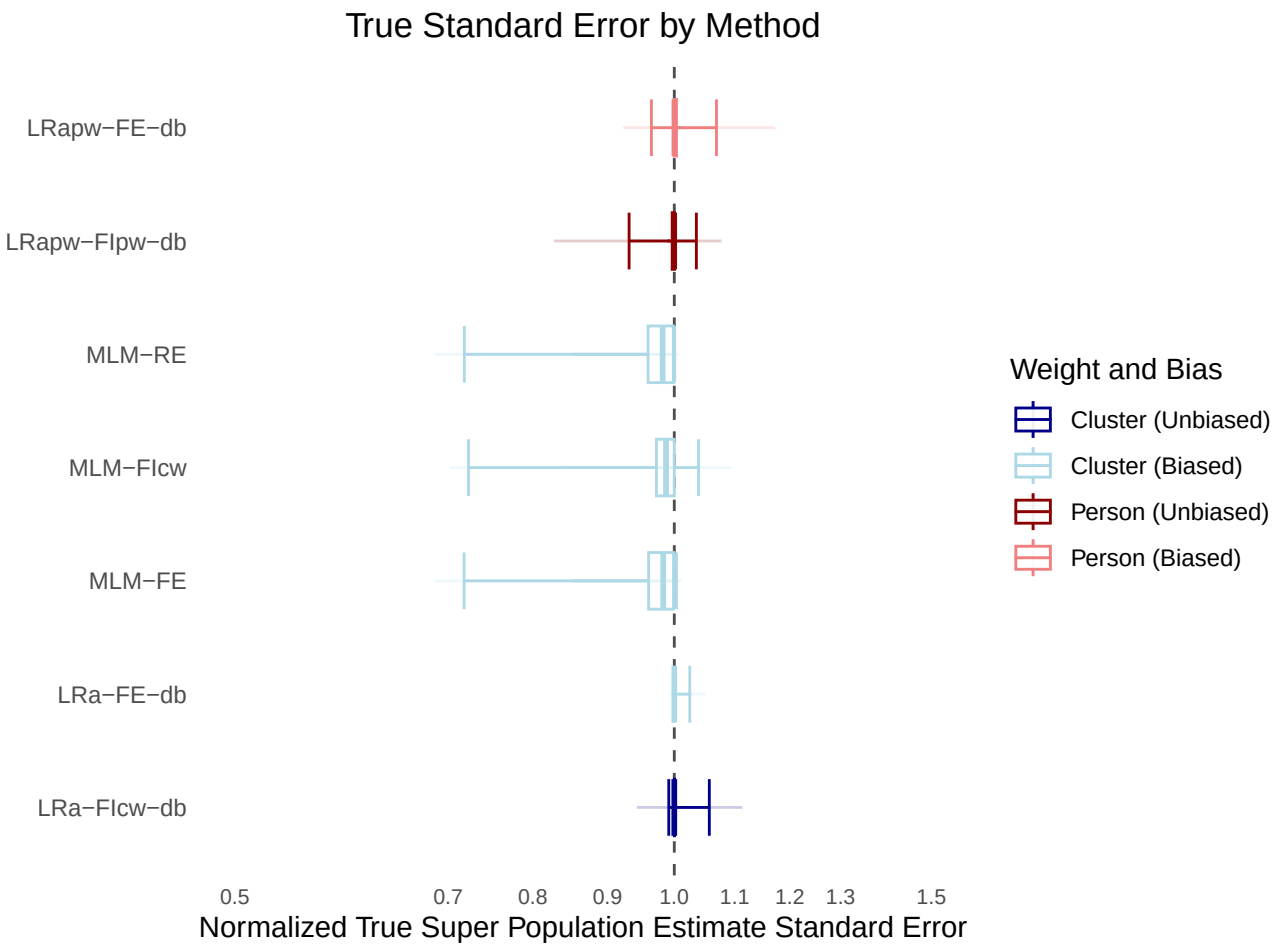


MLM−RE

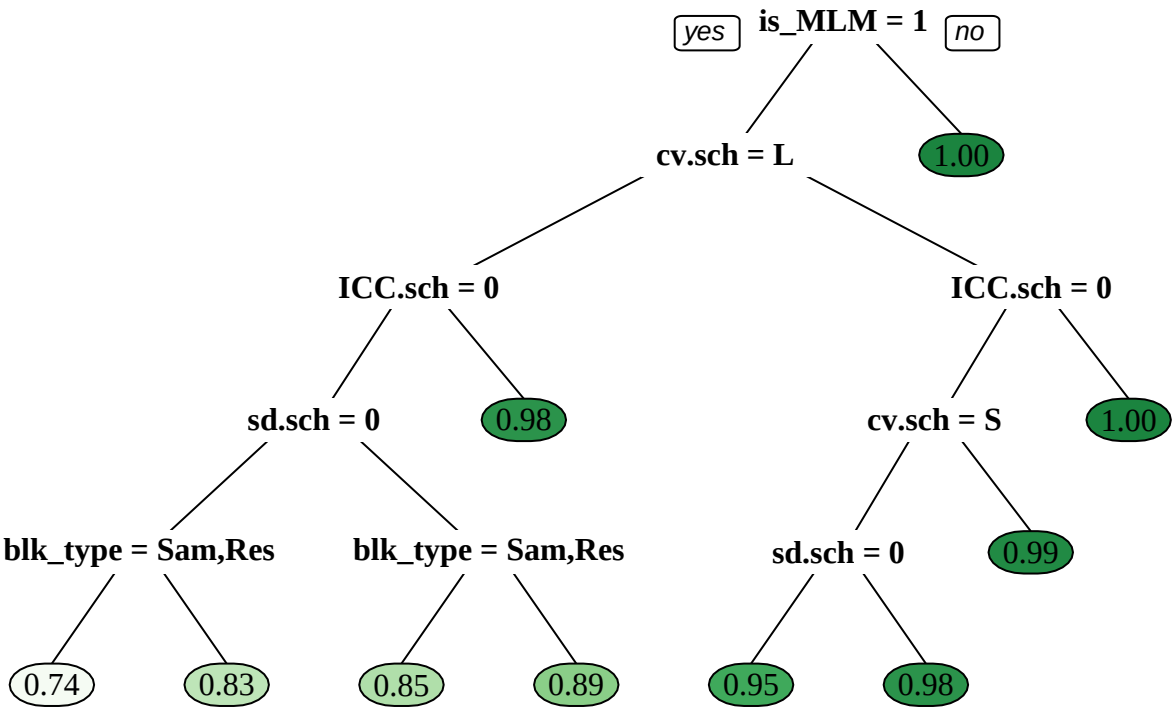


6 Are any of the estimators generally more precise than others? When?

In general all the different estimators have comparable precision (Figure 3). Excluding the multi-level estimators, within a given scenario and a targeted estimand (person weighted vs. cluster weighted) all the estimators have standard errors that are at most 10% larger than the median. The multi-level estimators become more precise when there is the combination is large variation in cluster size and the school ICC is zero, otherwise their precision is close to the other estimators (Figure 4).


True Standard Error by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
0.5
0.7
0.8
0.9
1.0
1.1
1.2
1.3
1.5
Normalized True Super Population Estimate Standard Error
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

Decision Tree for Super Population True Standard Error



#When in the low df scenario did it run at all

```
##
## FALSE  TRUE
##  2796   852
```

```
##      K
## J.bar   7  10  20
##     S   0   0 912
##     M   0   0 912
##     L 912 912   0
```

```
##      K
## J.bar  7 10
##     S  0  0
##     M  0  0
##     L 48  2
```

```
##      K
## J.bar 10 20
##     S  0 48
##     M  0 48
##     L 46  0
```

7 Which estimators produce the best estimates of the treatment effect standard error?

1) Assess, for each estimator, when the standard errors are systematically too high or too low

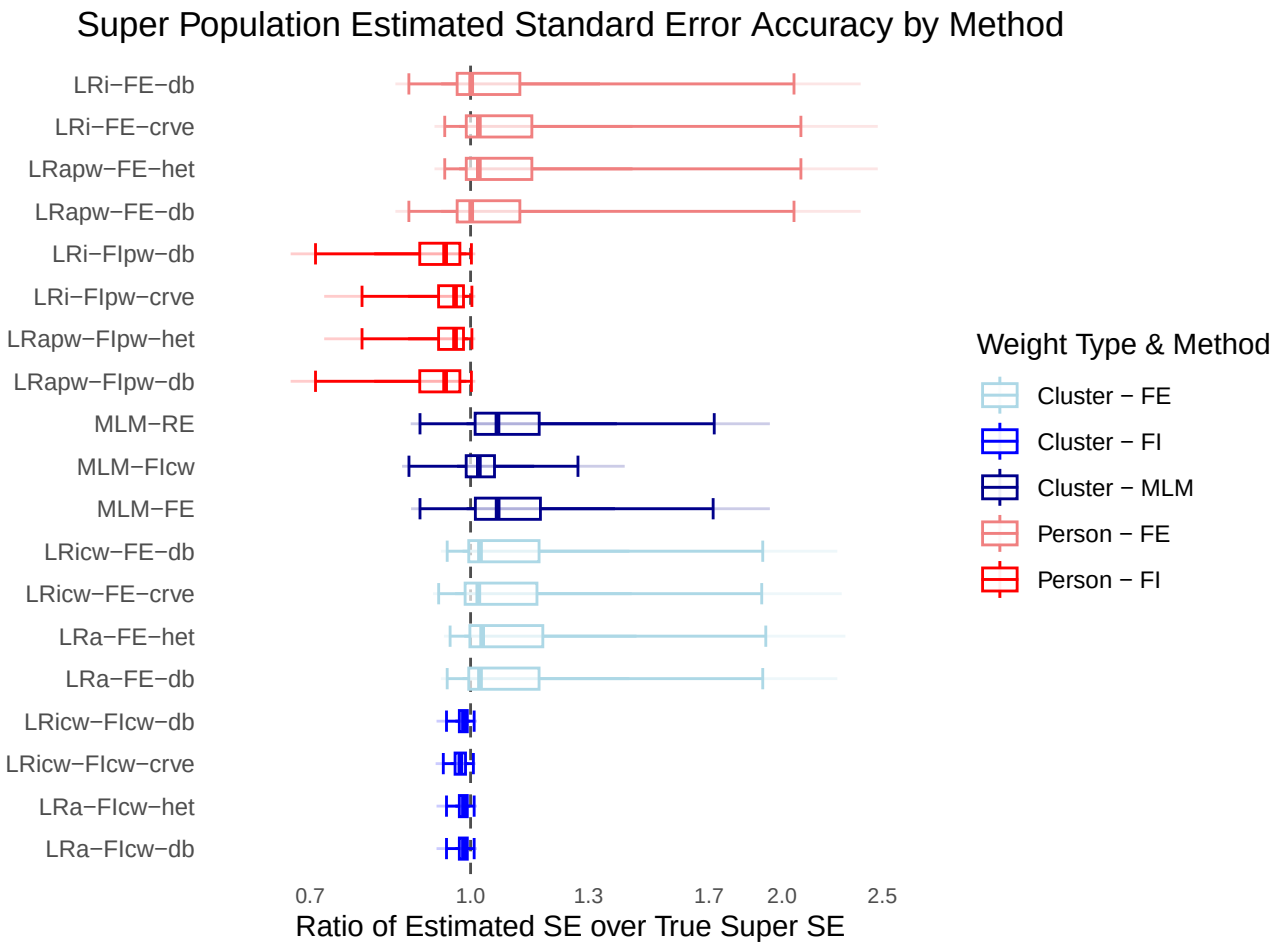
o These are all E[SE-hat] / SE, so all comparisons are internal to the estimator.

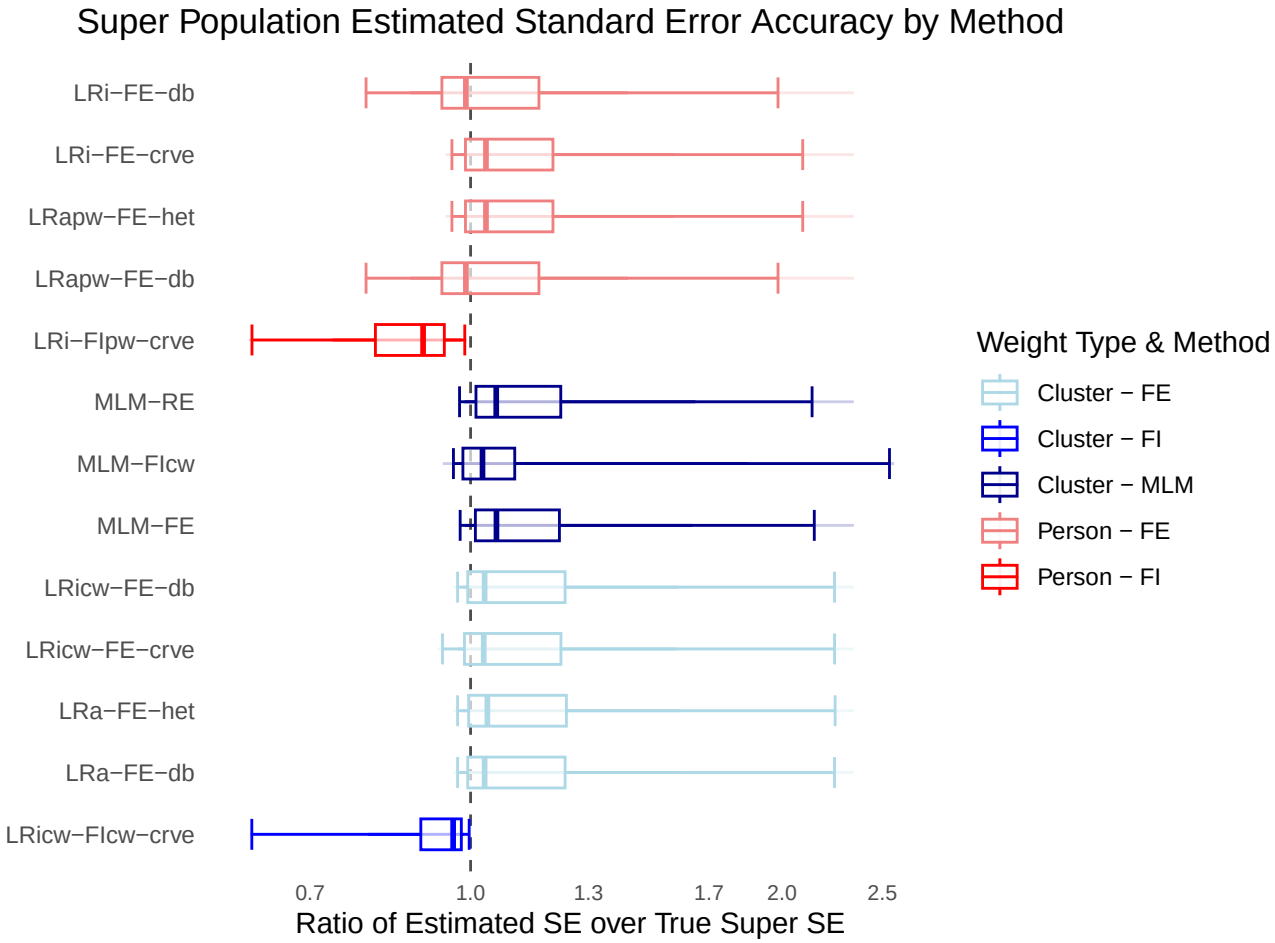
o Again identify boxplots of ranges, and then identify when an estimator has a long tail to being too low.

2) Assess, for each estimator, when is coverage poor for the estimators?

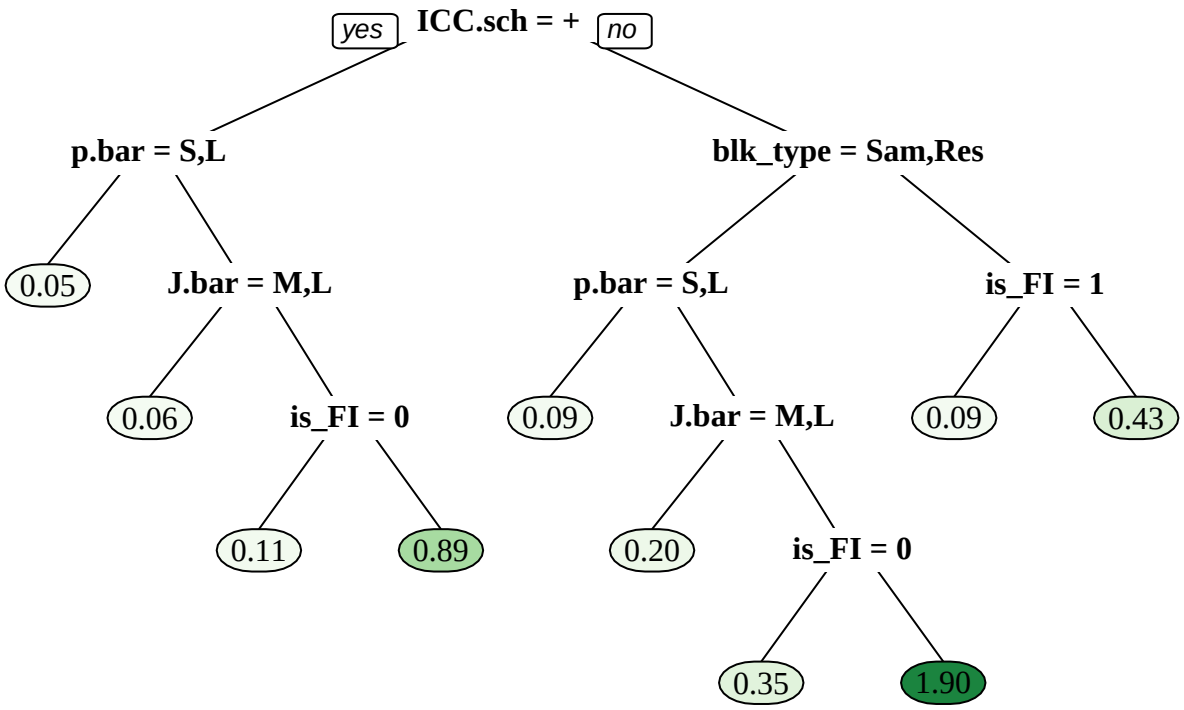
a. Calculate coverage for each scenario

b. Repeat the above.

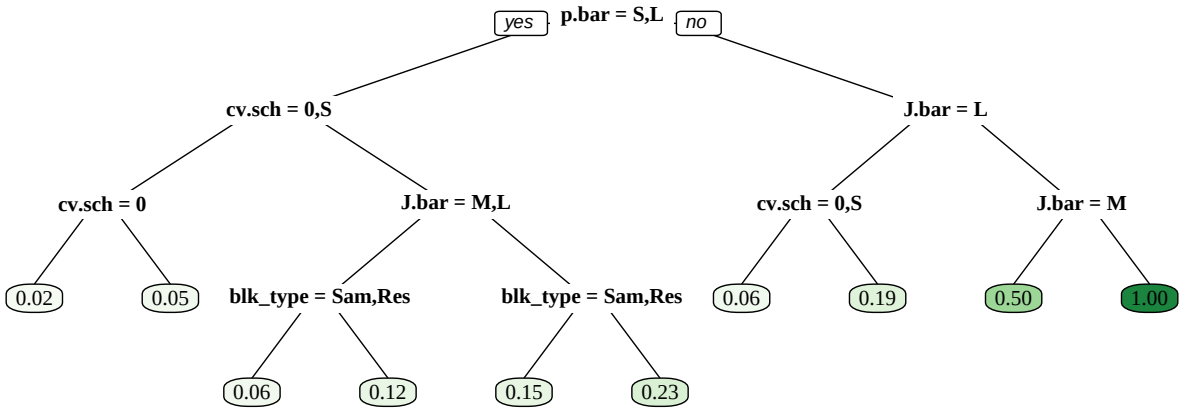



Super Population Estimated Standard Error Accuracy by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−crve
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−crve
0.7
1.0
1.3
1.7
2.0
2.5
Ratio of Estimated SE over True Super SE
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI

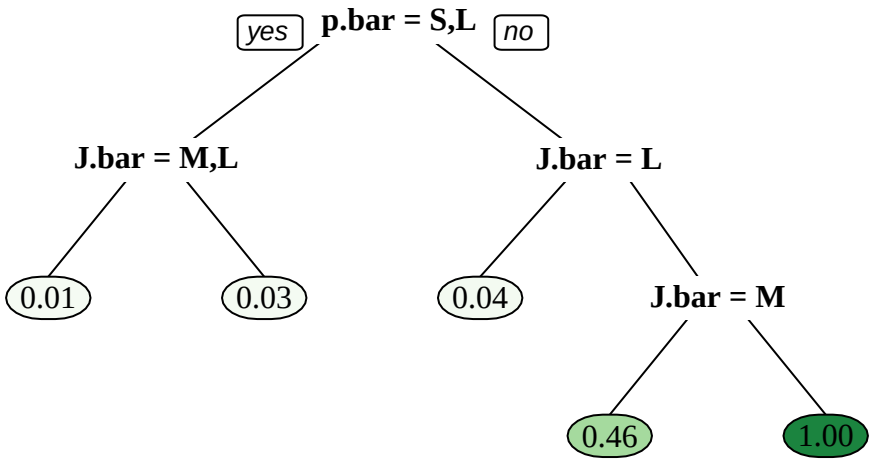
MLM


yes
ICC.sch = +
no
p.bar = S,L
blk_type = Sam,Res
0.05
J.bar = M,L
p.bar = S,L
is_FI = 1
0.06
is_FI = 0
0.09
J.bar = M,L
0.09
0.43
0.11
0.89
0.20
is_FI = 0
0.35
1.90

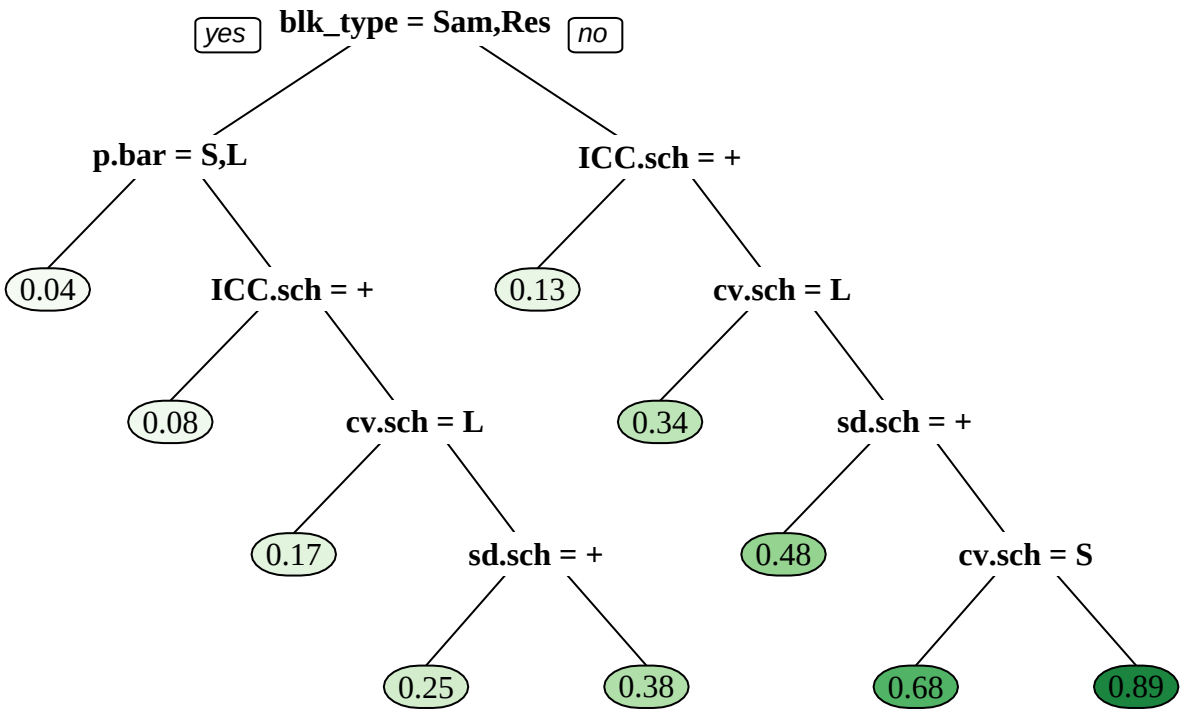
FI & PW


yes
p.bar = S,L
no
cv.sch = 0,S
J.bar = L
cv.sch = 0
J.bar = M,L
cv.sch = 0,S
J.bar = M
0.02
0.05
blk_type = Sam,Res
blk_type = Sam,Res
0.06
0.19
0.50
1.00
0.06
0.12
0.15
0.23

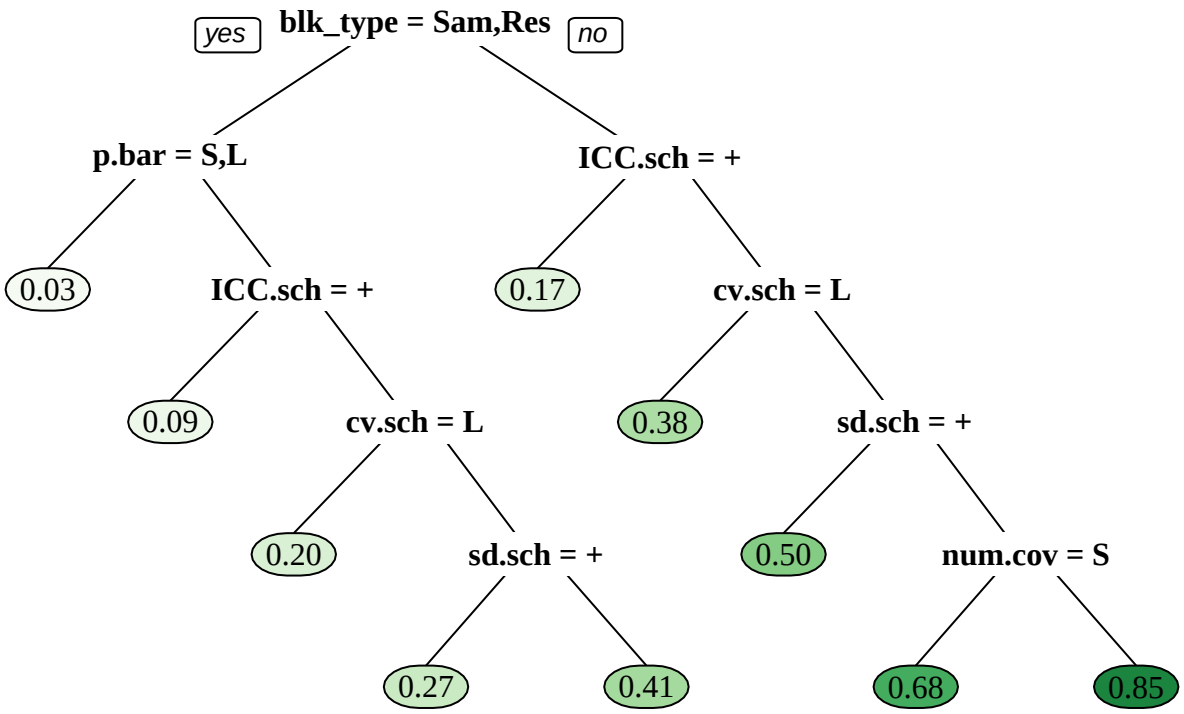
FI & CW

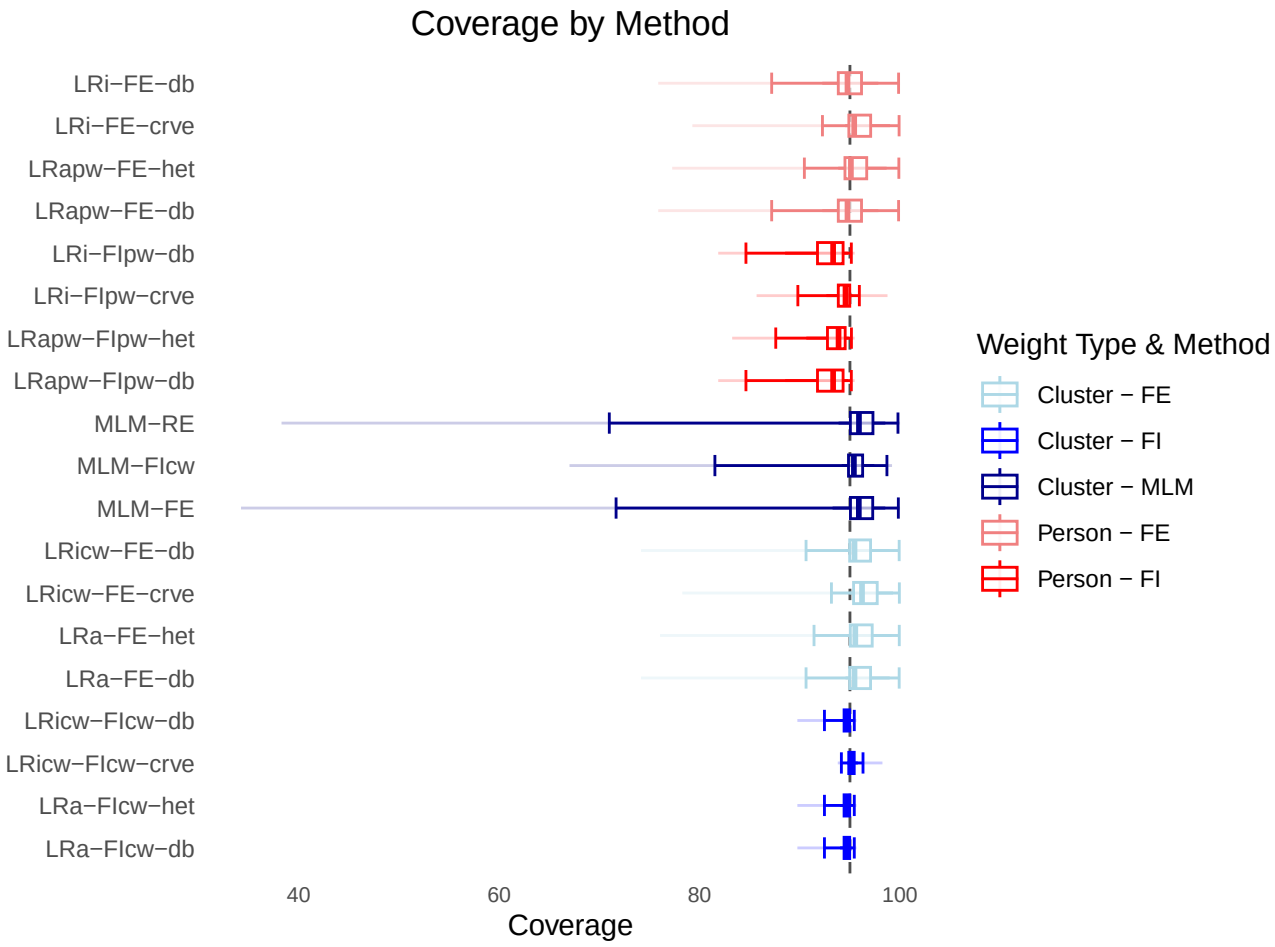


FE & PW

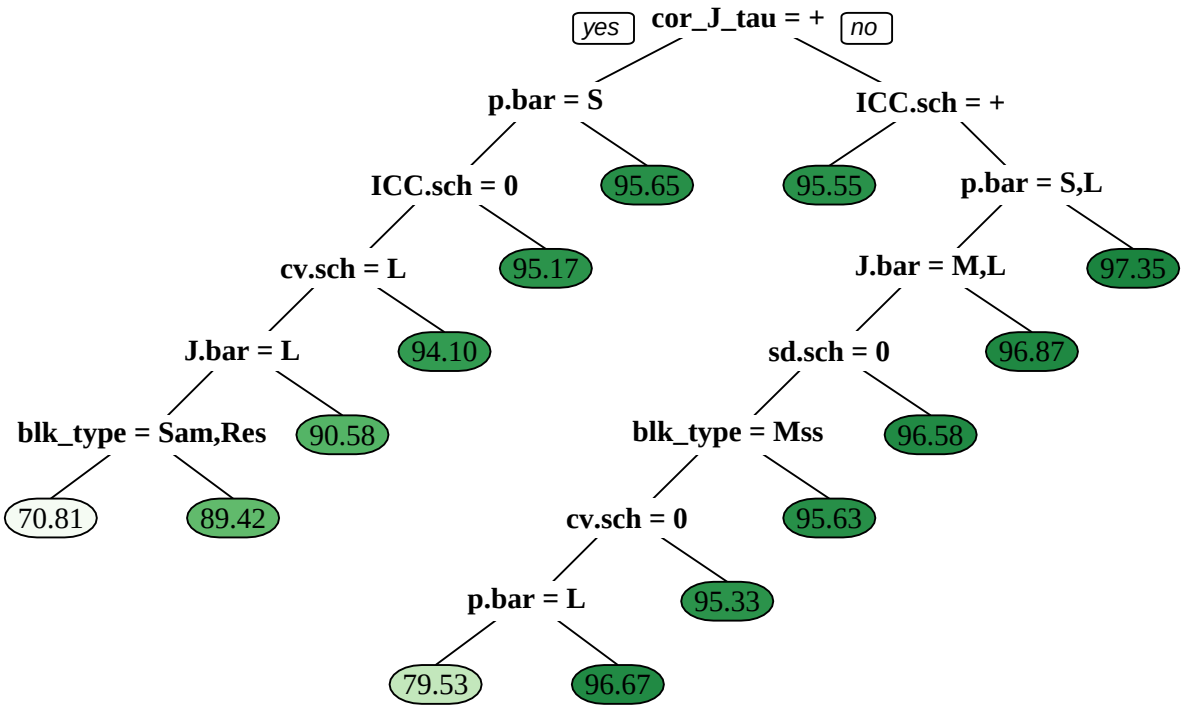

yes
blk_type = Sam,Res
no
p.bar = S,L
ICC.sch = +
0.04
ICC.sch = +
0.13
cv.sch = L
0.08
cv.sch = L
0.34
sd.sch = +
0.17
sd.sch = +
0.48
cv.sch = S
0.25
0.38
0.68
0.89

FE & CW


yes
blk_type = Sam,Res
no
p.bar = S,L
ICC.sch = +
0.03
ICC.sch = +
0.17
cv.sch = L
0.09
cv.sch = L
0.38
sd.sch = +
0.20
sd.sch = +
0.50
num.cov = S
0.27
0.41
0.68
0.85


Coverage by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
40
60
80
100
Coverage
Weight Type & Method
Cluster − FE
Cluster − FI
Cluster − MLM
Person − FE
Person − FI

MLM


yes
cor_J_tau = +
no
p.bar = S
ICC.sch = +
ICC.sch = 0
95.65
95.55
p.bar = S,L
cv.sch = L
95.17
J.bar = M,L
97.35
J.bar = L
94.10
sd.sch = 0
96.87
blk_type = Sam,Res
90.58
blk_type = Mss
96.58
70.81
89.42
cv.sch = 0
95.63
p.bar = L
95.33
79.53
96.67